

Speculative decoding breaks inference verification.

Coupling the provider's random numbers fixes it.

Jiawei Li · Inference Verification Gym (ivgym) · new measurements on one NVIDIA H100-80GB

Token-DiFR (Karvonen et al., 2025) audits an inference provider by replaying its returned tokens under seed-synchronised Gumbel-Max sampling — and assumes the provider decodes one token at a time. Real serving stacks speculate, and then DiFR flags every honest provider. This poster measures the break, the one-line fix, and how far the fix generalises.

≤ 0.022

false-positive effect size d' of the fixed scheme, all 12 swept settings

1.000

detection pAUC @ FPR ≤ 0.5% for int4 weights, in all 12 settings

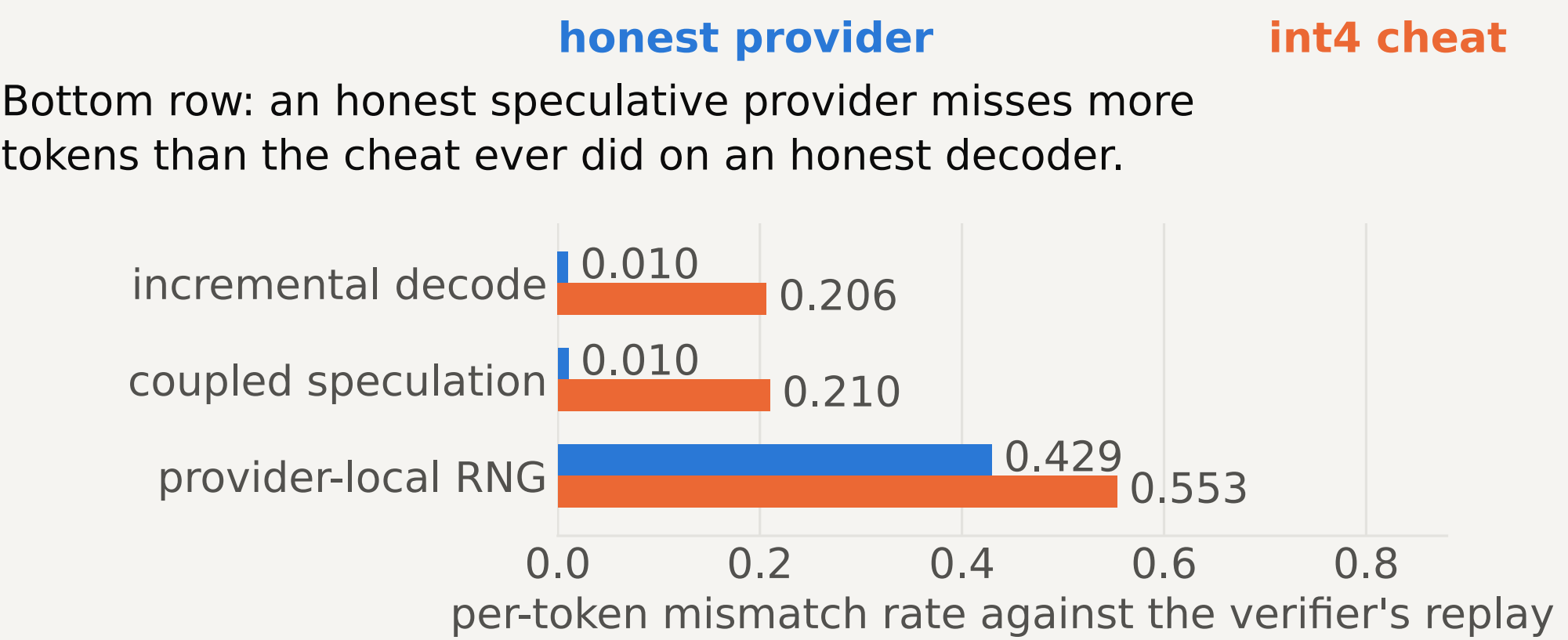
7.5%

of the speculative speedup is what auditability costs

1 • THE PROBLEM

The honest null collapses

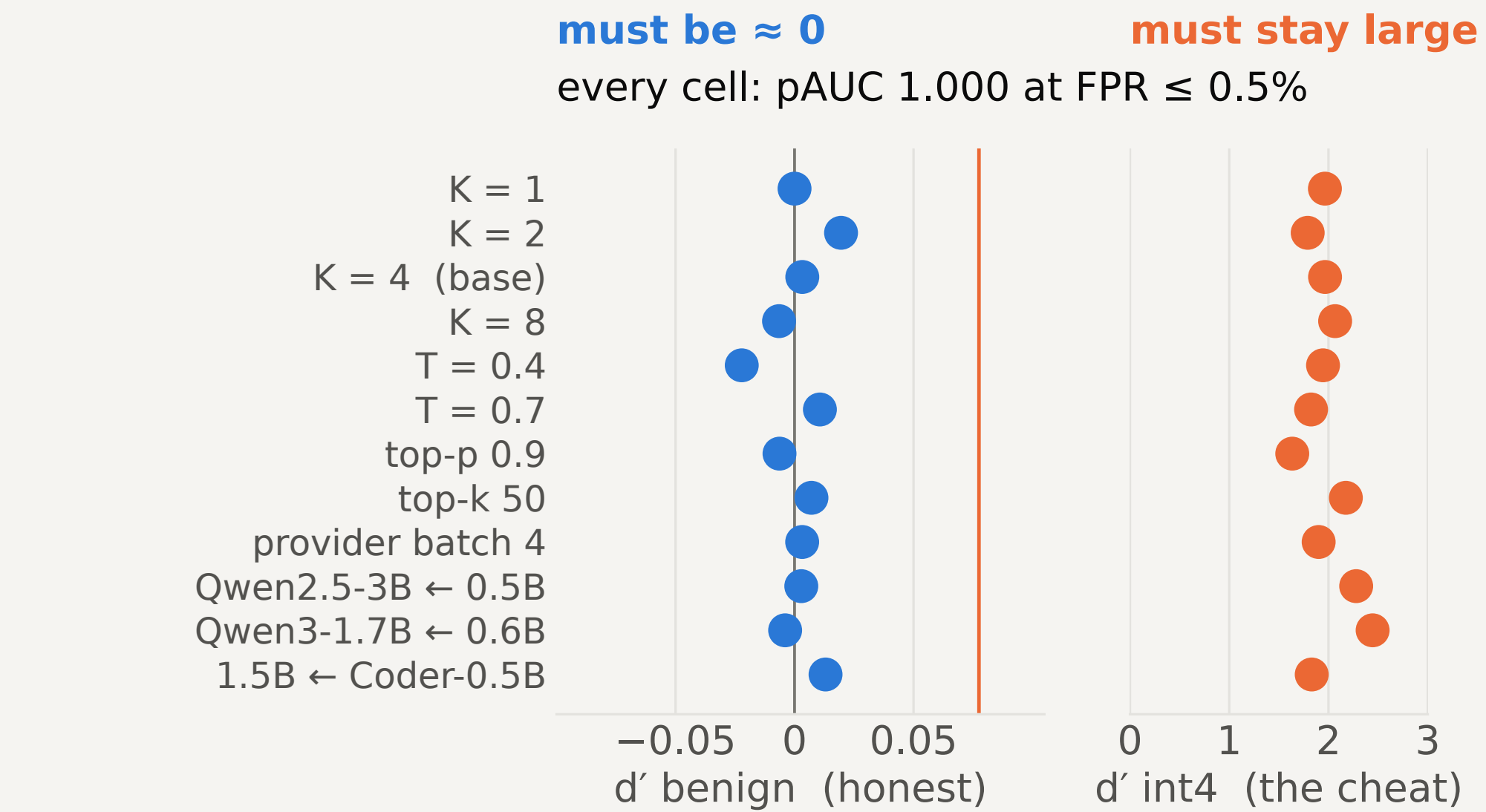
An honest provider reproduces the replayed token at ~99% of positions; a provider-local RNG destroys that null. Across four target/draft pairs the benign shift is d' 1.7–4.2, against 0.23–0.34 for an int4 cheat re-calibrated in the same regime.



4 • DOES IT GENERALISE?

The fix holds on every axis swept

Left: the false-positive side. Right: the power side — int4 weights against honest, in the same regime.



The orange rule marks $d' = 0.0775$, the repo's 4-bit-quantisation signal — 3–80× the benign shift the fix leaves behind.

2 • THE FIX: COMMON RANDOM NUMBERS

Couple the provider's Gumbels

Hand the draft the same Gumbel vector the verifier will use. The draft proposes

$$x_i = \arg \max_x [\log q(x)/T + G_{t+i}(x)]$$

and the target accepts it if and only if

$$x_i = \arg \max_x [\log p(x)/T + G_{t+i}(x)]$$

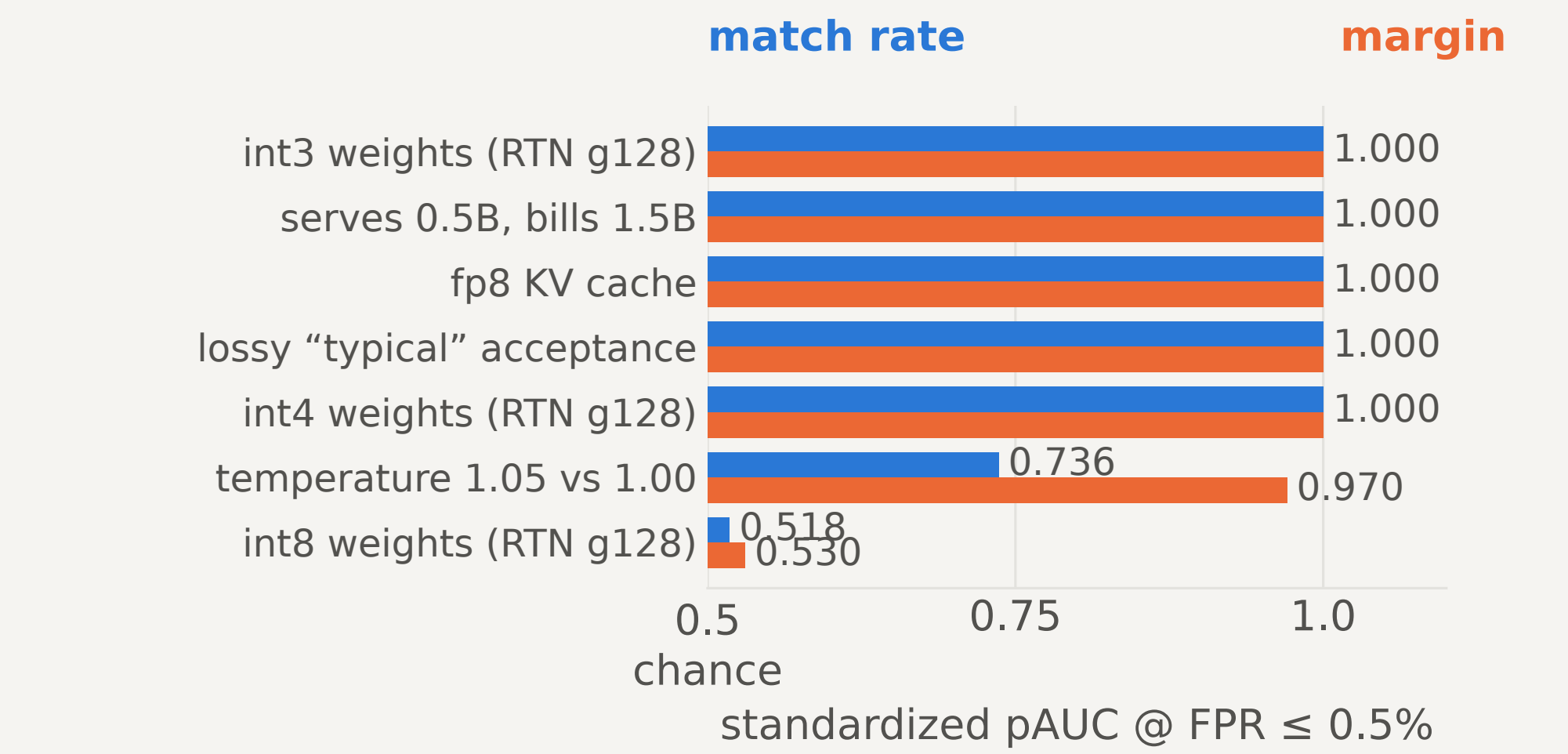
otherwise it emits its own argmax and the round ends. Still lossless in the Leviathan/Chen sense — but now every returned token is the one DiFR expects, and the verifier does not change at all.

Cost: 2.88 vs 3.11 tokens per target forward pass at K = 4 — 7.5% of the speculative speedup, and it still scales with depth (1.70 at K = 1 to 3.30 at K = 8).

5 • THE DEVIATION GRID

What the fix buys — and what it does not

Seven deviations under the fixed scheme, scored two ways: the binary match rate, and the post-Gumbel margin.



The margin dominates where a deviation changes how badly tokens miss, not whether they miss (0.736 vs 0.970 on the temperature retune). int8 is at chance on both.

3 • TAKEAWAYS

What we would deploy

- Audit the decoder, not just the weights.**
Speculation makes the calibration non-portable.
- Ask for coupled speculation, not for none.**
Output-identical to the reference decode, 7.5% slower.
- Aggregate the margin, not the match indicator.**
Except under top-p 0.9: 0.13% of honest tokens land outside the verifier's own nucleus, costing the margin 25× of its effect size.
- Neither of them fixes int8.**
Under the honest floor: a pool problem, not a seed problem.

6 • THE STRONGER STATEMENT

Not merely indistinguishable — identical

0.967

1.000

0.058

coupled speculation vs incremental decode

incremental decode rerun (the control)

provider-local RNG vs incremental decode

Fraction of returned tokens identical to the reference decode, same prompt and same seed. The coupled residual is one early numeric flip cascading through an autoregressive sequence.

Method: fp16 provider vs bf16 verifier on one H100 — a real cross-implementation numeric channel, not injected noise. 58 cells (4 speculation depths × 3 temperatures × 3 truncation rules × 2 provider batch shapes × 4 target/draft pairs × 8 provider deviations), 16 chat prompts × 192 tokens = 3 072 tokens per cell, scored at an 8.0% batch/pool ratio inside the repo's enforced 10% ceiling.
Reproduce: python -m experiments.exp_specdec_difr_sweep_gpu --out sweep · the same script with --identity · this poster: python paper/make_poster.py · artifacts: docs/results/specdec_difr_{sweep.jsonl, sweep_summary.json, identity.json, sweep.md}
References: Karvonen et al., DiFR: Inference Verification Despite Nondeterminism (2025) · Leviathan et al. (2022) and Chen et al. (2023), speculative decoding · code and every run artifact: the inference-verification (ivgym) repository.